

FIRE-RATED WINDOWS & PORTHOLES – A0 / A30 / A60

1. Product Description

- Product Type: Fire-rated marine window or porthole
- Fire Class Ratings: A0, A30, A60
- Compliance: Designed and tested according to ISO 5797:2004

2. Fire Resistance Classification

Class	Resistance Time	Application Area
A0	0 minutes	Accommodation, corridors
A30	30 minutes	Control rooms, machinery
A60	60 minutes	Bridge, engine room bulkhead

3. Standard Compliance and Certification

- ISO 5797 compliance
- Certified by Bureau Veritas
- Third-party testing report: YES

4. Material Specification

- Frame Material: Steel (Mild Steel / AISI 316L)
- **Glass Build-Up:**
 - Multi-layered laminated glass
 - Fire-rated interlayer (CFL ,CFL Lite Marine)
 - Total thickness: ~25mm to 30mm depending on class

5. Design & Construction

- Welded or bolted frame types
- Watertight sealing system
- We can fit with hinged deadlight/lid cover ort portable
- Surface Treatment : Sand Blasted - Marine Grade Prime, Topcoat

6. Thermal & Optical Properties

- Thermal insulation: Optional double-glazed
- Optional: Heat-resistant , UV-protection

7. Customization Options

- Heating elements (24V to 440V)
- Sun screens / blinds
- EMI/EMC shielding integration
- Bullet resistance (upon request)

8. Installation & Mounting

- Supplied fully assembled, tested
- Pre-drilled holes / welding brackets
- Installation manual and maintenance instructions included

9. Maximum Clear Opening Dimensions (without Mullion/Division):

- A0 & A30 Fire-Rated Windows: up to 2230mm x 2400mm
- A60 Fire-Rated Windows: up to 2104mm x 1804mm

These sizes can be supported without the need for intermediate mullions, providing wide, unobstructed viewing areas while maintaining certified fire protection.

Safety ;

Fire-rated marine windows and portholes are a critical safety component in vessel design.

They are engineered to withstand high temperatures for specific durations (A0, A30, A60) according to ISO 5797 and IMO FTP Code requirements. These windows:

- Prevent the spread of fire between compartments
- Maintain structural integrity during fire exposure
- Protect evacuation and control areas, such as the wheelhouse or escape routes
- Support overall compliance with SOLAS regulations and classification society rules

By incorporating fire-rated windows in designated zones, shipbuilders significantly increase onboard safety and survivability in emergency situations.